Unit 4 Progress Check: FRQ Part B

1. The GameAccount class is used to store information about players of an online game. A partial declaration of the GameAccount class is shown.

```
public class GameAccount
{
  /**
   * Returns the username of the player
   */
  public String getUsername()
  { /* implementation not shown */ }

  /**
   * Returns the age of the player
   */
  public int getAge()
  { /* implementation not shown */ }

  /* There may be instance variables, constructors,
     and methods that are not shown. */
}
```

The GamePlatform class maintains an ArrayList of GameAccount objects named allPlayers. A partial declaration of the GamePlatform class is shown.

```
public class GamePlatform
{
  /** An ArrayList of all player accounts */
  public ArrayList<GameAccount> allPlayers;

  /**
   * Returns the average age of players in allPlayers whose age
   * is between 13 and 19, inclusive
   * Preconditions: allPlayers contains at least one player
   *                whose age is between 13 and 19.
   *                allPlayers is not null and contains no null
   *                elements.
   * Postcondition: allPlayers is unchanged.
   */
  public double avgTeenAccounts()
  { /* to be implemented */ }

  /* There may be instance variables, constructors,
     and methods that are not shown. */
}
```

Write the GamePlatform method avgTeenAccounts. The method should return the average age of the players in allPlayers whose age is between 13 and 19, inclusive.

For example, suppose allPlayers has the following contents:

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Username	"star_guy"	"a_knight"	"dino_dan"	"cool_cat"	"happy_hero"	"fox_sox"
Age	23	14	12	15	13	19

For these contents of allPlayers, the method avgTeenAccounts should return 15.25. There are four players whose age is between 13 and 19, inclusive ("a_knight", "cool_cat", "happy_hero", and "fox_sox"). The average age of these four players is $(14 + 15 + 13 + 19)/4$, which is 15.25.

Complete method avgTeenAccounts.

```
/**
 * Returns the average age of players in allPlayers whose age
 * is between 13 and 19, inclusive
 * Preconditions: allPlayers contains at least one player
 *                whose age is between 13 and 19.
 *                allPlayers is not null and contains no null
 *                elements.
 * Postcondition: allPlayers is unchanged.
 */
public double avgTeenAccounts()
```

2. The Game class is used to store the scores of a home team and an away team in a game. A partial declaration of the Game class is shown.

```
public class Game
{
  /**
   * Returns the home team's score for this game
   */
  public int getHomeScore()
  { /* implementation not shown */ }

  /**
   * Returns the away team's score for this game
   */
  public int getAwayScore()
  { /* implementation not shown */ }

  /* There may be instance variables, constructors,
     and methods that are not shown. */
}
```

The Season class maintains an ArrayList of Game objects named allGames. A partial declaration of the Season class is shown.

```
public class Season
{
```

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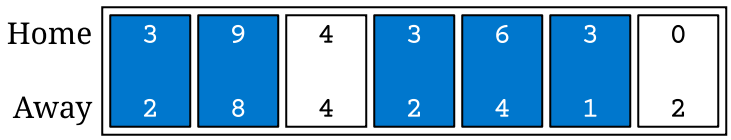
```
  /** The list of all games played during the season */
  private ArrayList<Game> allGames;

  /**
   * Returns the number of games in the longest streak of
   * consecutive games where the home team scored more than
   * the away team
   * Preconditions: allGames is not empty.
   *                No elements of allGames are null.
   * Postcondition: allGames is unchanged.
   */
  public int getLongestHomeWinStreak()
  { /* to be implemented */ }

  /* There may be instance variables, constructors,
     and methods that are not shown. */
}
```

Write the Season method getLongestHomeWinStreak. The method should return the number of games that make up the longest winning streak by the home team. A winning streak is a number of consecutive games won by a team. A game is considered a win for the home team if the home team score is greater than the away team score.

For example, suppose allGames contains the following Game objects. The shaded elements represent home team wins.



For these contents of allGames, the method getLongestHomeWinStreak() should return 3 because the longest winning streak for the home team is 3 games.

Complete method getLongestHomeWinStreak.

```
/**
 * Returns the number of games in the longest streak of
 * consecutive games where the home team scored more than
 * the away team
 * Preconditions: allGames is not empty.
 *                No elements of allGames are null.
 * Postcondition: allGames is unchanged.
 */
public int getLongestHomeWinStreak()
```